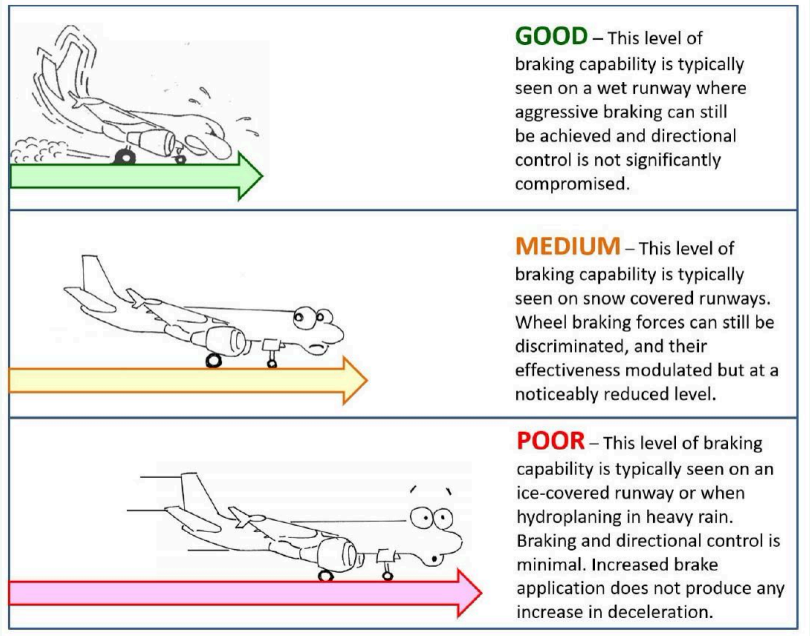
MODULE 4 · RUNWAYS AND PERFORMANCE

Lesson 15: Runway Condition Reporting: GRF, RCAM and RWYCC

Lesson 15 of 30 · about 10 min

LEARNING OBJECTIVES

- Explain the ICAO Global Reporting Format (GRF) and the Runway Condition Report (RCR).
- Interpret runway condition codes (RWYCC) using the RCAM.
- Make and use pilot braking action reports correctly.



Braking action terms match the runway condition codes: good, medium, poor and the steps in between.

The Global Reporting Format

Since November 2021, ICAO's **Global Reporting Format (GRF)** has been the standard method for assessing and reporting runway surface conditions. Its purpose is to give pilots a common language that links directly to aircraft performance data. The FAA uses an equivalent system called TALPA.

The aerodrome operator assesses the runway and issues a **Runway Condition Report (RCR)**, which reaches crews through ATIS, ATC and SNOWTAM.

Reading a Runway Condition Report

The aircraft performance part of the RCR is given for each **third** of the runway, in the direction of the **lower** runway designator:

```
XXXX 01170830 09 5/5/2 100/100/50 NR/NR/06 WET/WET/SLUSH
```

Element	Example	Meaning

Aerodrome / date-time	`XXXX 01170830`	Location and time of assessment (month, day, hour, minute UTC)
Runway	`09`	Lower runway designator
RWYCC per third	`5/5/2`	Runway condition code for each third
Coverage per third	`100/100/50`	Percentage of each third covered (reported as 25, 50, 75 or 100)
Depth per third	`NR/NR/06`	Depth in mm of loose contaminant (water, slush, dry or wet snow); NR = not reported
Contaminant per third	`WET/WET/SLUSH`	Type of surface condition



Example report `5/5/2`: runway condition codes for each third, given in the direction of the lower runway designator.

The Runway Condition Assessment Matrix (RCAM)

RWYCC	Runway surface description	Pilot braking action
6	DRY	—
5	FROST; WET (including damp, and 3 mm of water or less); SLUSH, DRY SNOW or WET SNOW 3 mm or less	Good
4	COMPACTED SNOW at OAT −15 °C and below	Good to medium
3	WET (“slippery wet” runway); DRY SNOW or WET SNOW (any depth) ON TOP OF COMPACTED SNOW; DRY SNOW or WET SNOW more than 3 mm; COMPACTED SNOW at OAT above −15 °C	Medium
2	STANDING WATER more than 3 mm; SLUSH more than 3 mm	Medium to poor
1	ICE	Poor
0	WET ICE; WATER ON TOP OF COMPACTED SNOW; DRY SNOW or WET SNOW ON TOP OF ICE	Less than poor

The aerodrome may upgrade or downgrade a code using measurements, observations and pilot reports. A runway with RWYCC 0 is not used for take-off or landing.

Pilot braking action reports

Pilots report braking action using the same terms — **good, good to medium, medium, medium to poor, poor, less than poor** — for example: “Braking action medium to poor, runway 27, second half.”

- Report whenever braking or directional control is worse than reported, or when asked.
- Your report is an important input to the aerodrome's assessment and to the crew behind you.
- Reports age quickly in active precipitation. Always check the time of the RCR and the trend.

KEY TAKEAWAYS

- RCR is reported per runway third, in the direction of the lower designator.
- RWYCC 6 = dry … 0 = less than poor. 2 = standing water or slush more than 3 mm; 1 = ice.
- Use the reported codes directly with the performance data.
- Report braking action that is worse than reported.

KNOWLEDGE CHECK

A runway third is covered with 6 mm of slush. What RWYCC would normally be reported?

A. 5

B. 3

C. 2

D. 0

Answer: C. In the RCAM, standing water or slush deeper than 3 mm corresponds to RWYCC 2 (medium to poor).

Training material only. It does not replace the Aircraft Flight Manual, FCOM, QRH, MEL or the Domestic Airlines Operations Manual, which always take precedence.